

ORIGINAL ARTICLE

WILEY

Emissions of VOCs, SVOCs, and mold during the construction process: Contribution to indoor air quality and future occupants' exposure

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Funding information

Agence de l'Environnement et de la Maîtrise de l'Energie, Grant/Award Number: 1504C0005

Abstract

Building materials and human activities are important sources of contamination indoors, but little information is available regarding contamination during construction process which could persist during the whole life of buildings. In this study, six construction stages on two construction sites were investigated regarding the emissions of 43 volatile organic compounds (VOCs), 46 semi-volatile organic compounds (SVOCs), and the presence of 4 genera of mold. Results show that the future indoor air quality does not only depend on the emissions of each building product but that it is also closely related to the whole implementation process. Mold spore measurements can reach 1400 CFU/m³, which is particularly high compared with the concentrations usually measured in indoor environments. Relatively low concentrations of VOCs were observed, in relation to the use of low emissive materials. Among SVOCs analyzed, some phthalates, permethrin, and hydrocarbons were found in significant concentrations upon the delivery of building as well as triclosan, suspected to be endocrine disruptor, and yet prohibited in the treatment of materials and construction since 2014. As some regulations exist for VOC emissions, it is necessary to implement them for SVOCs due to their toxicity.

KEYWORDS

construction process, dwellings, mold, offices, SVOC, VOC

1 | INTRODUCTION

People spend about 90% of their time in closed environments, and most of this time occurs at home.^{1,2} The Observatory on Indoor Air Quality (OQAI) observed that indoor pollution is different from outdoor pollution, especially in terms of the presence of some substances that are not detected outdoors or by higher concentrations indoors. In a national survey including 567 French dwellings, it has been demonstrated that the pollutants investigated were present

at quantifiable levels in most French dwellings and included physical, chemical, and microbiological compounds.³ Other studies found higher concentrations of pollutants indoors than outdoors, measuring volatile organic compounds (VOCs)^{4,5} or semi-volatile organic compounds (SVOCs).^{6,7} High mold concentrations are also often detected in dwellings.^{8,9}

Pollutants found in indoor environments have or are suspected to be related to health adverse effects. VOCs are known to cause respiratory health effects such as asthma or rhinitis¹⁰ and are

suspected to involve the development of severe diseases as cancer after a long period of inhalation exposure.¹¹ Formaldehyde is an example of substance classified as carcinogenic for humans by the International Agency for Research on Cancer (IARC).¹² SVOCs are suspected to have reprotoxic or neurotoxic effects.^{13,14} They are emitted from various sources in living environments¹⁵ and include a wide range of molecules with different properties: pesticides, biocides, plasticizers, flame retardants, surfactants, and lubricants used as active substances or additives. They are present in the gas phase, airborne particles, and settled dust¹⁶; therefore, humans are exposed through different pathways including inhalation, ingestion, and dermal contact.¹⁷ Concerning fungal contamination, the presence of mold in dwellings is correlated with the onset of asthma in children and adults.¹⁸⁻²¹

These agents or contamination come from different sources including building materials, furnishings, cleaning supplies, and human activities. High VOC concentrations have often been observed in newly built or renovated residential buildings.²²⁻²⁴ Nevertheless, little information is available regarding the contamination during the construction process which could persist during the whole life of buildings. Even if some authors such as Mendell et al²⁵ have alerted the international research community to the risk of contamination of workers during the construction phase, we have focused our study on the potential health effects on future occupants. To our knowledge, only six international publications are available regarding VOC contamination during construction processes in the scope of general population and public health exposure. Liu et al²⁶ found a significant VOC contamination during the construction and finishing of a new museum in China and suggested that potential contaminant sources were various kinds of wall paints and sprayed glass cleaning agents. Other authors found that the introduction of furniture during the construction process was responsible for emissions of the largest amount of pollutants, including toluene and formaldehyde.^{27,28} Liang et al measured the VOC concentrations during five construction stages of a new apartment and identified the sources of different compounds. Toluene and alpha-pinene increased considerably after the door and doorframe were installed. Likewise, propylene glycol increased after application of the wall paint and benzene after introduction of some furniture.²⁹ Ochs et al³⁰ identified that the door varnishing was responsible for a high emission of butanone and other carbonyl compounds. Plaisance et al³¹ measured the emissions of VOCs at six construction stages in three energy-efficient timber frame houses and found that high m,p-xylenes and ethylbenzene concentrations at the time of the structural work were due to the emission from the polyurethane adhesive mastic used as a sealing material. They also found that a large number of materials were sources of aldehydes which became the highest proportion in the chemical composition in the last stages of construction. VOC emissions from building materials decrease with time.³²⁻³⁴ Therefore, occupants will be exposed to VOCs emitted by building materials during few months to several years after interior construction was completed, with, for example, secondary aldehyde emissions due

Practical Implications

- The 2012 French Thermal regulation enforces requirements on energy consumption and air permeability of the built envelope.
- These economic requirements could have consequences on indoor air quality by increasing the concentration levels of pollutants from building materials if they are not associated with an increase in ventilation rates.
- Results of this study raise questions about indoor air quality management from the early stages of the construction project, which could reduce the indoor air pollution at delivery.

to ozone reactions on building materials.³⁵ Regarding fungal contamination, Sautour et al³⁶ conducted a survey in hospital during a period of building construction and found that there were increases in airborne fungal spores at the beginning of construction work and that the most frequently recovered airborne fungi were *Penicillium* spp., *Aspergillus* spp., and *Bjerkandera adusta*. Emissions of mold that have developed during the construction process may persist throughout the life of the building if mold growth occurred on non-visible materials, as drywall.^{37,38} These molds, even hidden at the delivery phase, could reappear during the occupation of buildings or remain active behind a lining wall, and could be correlated with the onset of asthma. No studies were found regarding SVOC emissions during construction stages while SVOCs persist several years after the end of the construction process or throughout the life of the building and contribute to the long-term exposure of occupants.^{15,39}

In France, some regulations exist regarding the emissions of construction materials. A extremely low emission factor ($1 \mu\text{g}/\text{m}^3$) has been set for four priority substances (trichloroethylene, benzene, bis(2-ethylhexyl)phthalate, and dibutyl phthalate) in order to prevent the producers to add these compounds in their products.⁴⁰ A labeling has also been introduced regarding indoor air emissions from construction and decoration products.⁴¹ This labeling indicates the level of VOC emissions, allowing a choice of products that are less harmful to health. Some indoor air quality guidelines (IAGV) exist for some VOCs, but no values are available for SVOCs. Moreover, no regulations exist regarding the maximum values of pollutants allowed in the indoor air at delivery of buildings.

This project focuses on the impact of the different trades (painter, plasterer, mason, etc) considering differences in process (materials, implementation conditions) and implementation of ventilation equipment to better document the impact of the different construction stages on the air quality at delivery of a building. It is the first time in France that a project aims to measure as many parameters during the different construction stages: VOCs, SVOCs, and mold. The objective is to identify the potential sources of pollution that can impact the indoor air quality at the delivery of a building and

TABLE 1 Details of the measurement campaigns realized in air and dust

Construction stage	Construction site	Measurement campaigns	Analyzed parameters	Measurement matrix	Location	Sampling duration
Initial state	A	February 22 to 26, 2016	TVOCs	Air	A1	Continuously
			VOCs	Air	A0, A1	5 days
	B	January 10, 2017	TVOCs	Air	B2	Continuously
			VOCs	Air	B0, B1, B2	6.5 hours
			Mold	Air	Outdoor, B0, B1, B2	30 seconds
Preparation of supports	A	April 4, 2016	TVOCs	Air	A1	Continuously
			Mold	Air	Outdoor, A0, A1	30 seconds
		April 18 to 22, 2016	TVOCs	Air	A1	Continuously
			VOCs	Air	A0, A1	5 days
	B	February 28, 2017	TVOCs	Air	B2	Continuously
			Mold	Air	Outdoor, B0, B1, B2	30 seconds
		April 13, 2017	TVOCs	Air	B2	Continuously
			VOCs	Air	B0, B1, B2	6.5 hours
		April 14, 2017	TVOCs	Air	B2	Continuously
			Mold	Air	Outdoor, B0, B1, B2	30 seconds
Finish paints	A	May 13, 2016	TVOCs	Air	A1	Continuously
			VOCs (except aldehydes)	Air	A0, A1	180 minutes
			Mold	Air	Outdoor, A0, A1	30 seconds
			SVOCs	Dust from ventilation ducts	Ground floor, first floor	—
			Mold	Dust from ventilation ducts	Ground floor, first floor	—
	B	May 25, 2016	TVOCs	Air	A1	Continuously
			Aldehydes	Air	A0, A1	390 minutes
		May 2, 2017	TVOCs	Air	B2	Continuously
			VOCs	Air	B0, B1, B2	6.5 hours
Soft floors	A	June 3, 2016	TVOCs	Air	A1	Continuously
			VOCs	Air	A0	6.5 hours
		June 8, 2016	TVOCs	Air	A1	Continuously
			VOCs	Air	A1	6.5 hours
	B	August 4 to 7, 2017	TVOCs	Air	B2	Continuously
			VOCs	Air	B0, B0 living room ^a , B2	6.5 hours
			Mold	Air	Outdoor, B0, B0 living room, B2	30 seconds
			SVOCs	Airborne PM ₁₀	B2	96 hours
Last finishing works	A	June 21 to 24, 2016	TVOCs	Air	A1	Continuously
			VOCs	Air	A0, A1	6.5 hours
			SVOCs	Airborne PM ₁₀	A1	72 hours
	B	September 19, 2017	TVOCs	Air	B2	Continuously
			VOCs	Air	B0, B1, B2	6.5 hours
			Mold	Air	Outdoor, B0, B1, B2	30 seconds

(Continues)

TABLE 1 (Continued)

Construction stage	Construction site	Measurement campaigns	Analyzed parameters	Measurement matrix	Location	Sampling duration
Delivery	A	July 27, 2016	TVOCs	Air	A1	Continuously
			VOCs	Air	A0, A1	6.5 hours
	B	October 3 to 6, 2017	TVOCs	Air	B2	Continuously
			VOCs	Air	B0, B1, B2	6.5 hours
			Mold	Air	Outdoor, B0, B1, B2	30 seconds
			SVOCs	Airborne PM ₁₀	B2	96 hours

Abbreviations: SVOCs, semi-volatile organic compounds; TVOCs, total volatile organic compounds; VOCs, volatile organic compounds.

^aMeasurement in the living room on the ground floor.

thus contribute to the exposure of the future occupants to harmful contaminants.

ground floor instead of B1, to study the impact of the storage of soft floors on the VOC emissions.

2 | MATERIALS AND MEASUREMENT DESIGN

2.1 | Construction sites

Two sites of new construction operations were investigated: an office building and a housing building. The two sites were located in the west part of France (Nantes) in a temperate oceanic climate.

Site A consists of the construction of a new office building. The building, with a net internal area (NIA) of 505 m², is built on two levels: a ground floor and a first floor. The building is based on concrete sails cast in place for vertical structure and prestressed slabs and sail cast in place at the top (including for the high floor) for horizontal structure. Site A was poorly ventilated (few window openings), and a dual-flow ventilation was started just before the last measurement campaign, on July 25, 2016.

Site B consists of the construction of 38 dwellings spread over a set of five buildings. Measurements were carried out in a single building (n°2) with a NIA of 600 m². The building comprises a total of six dwellings spread over three levels: ground floor, first floor, and second floor. The construction process includes concrete sails cast in place, prefabricated concrete floors, and wood frame facades. The gables in concrete veil benefit from an external insulation and the wall wood frame of a distributed insulation. Site B was more ventilated than site A (window openings), and mechanical ventilation was not used during the measurement campaigns. Contrary to site A, smoking was observed during each visit (principally by the paint company), while this is normally prohibited on construction sites.

Two locations were chosen for the measurements on site A: room A0 on the ground floor and located at the Nord corner of the building and room A1 on the first floor and facing southeast.

Measurements on site B were conducted at three locations: bedroom B0 on the ground floor at the southeast corner, living room B1 on the first floor oriented at the northwest, and bedroom B2 on the second floor at the southeast corner. During the implementation of soft floors, measurements were done on the living room on the

2.2 | Building materials and energy performance

The energy performance of buildings was regulatory according to the 2012 French Thermal regulation.⁴² Measurements of air permeability were carried out using a blower door in accordance with the NF EN 13829 standard. The air permeability of the building envelope was equal to 0.61 m³/h.m² for site A (measured on March 2, 2016) and equal to 0.80 m³/h.m² for site B (measured at the end of construction), under 4 Pa.

All materials and products used have low emissivity with respect to compulsory labeling (class A + for most of materials and finishing products).⁴¹

2.3 | Construction stages and interest parameters

Six construction stages were investigated:

1. After the structural work: initial state (walls, roof, exterior joinery, building airtight)
2. During preparation of supports: application of coating and primer paint, implementation of the screed, installation of partitions and linings
3. During the application of finish paints
4. During the implementation of soft floors
5. During the last finishing works: finishing touches, setting up of wood interior joinery, before cleaning
6. At the delivery

Details of the measurement campaigns are presented in Table 1. Site A was investigated from February 22, 2016, to July 27, 2016. Site B was investigated from January 10, 2017, to October 6, 2017. Regarding VOCs, 30 measurements were carried out in total on both sites: one measurement at each construction stage in each of the five rooms investigated. Regarding SVOCs, four measurements in total of both sites were carried out: one measurement on the dust of ventilation ducts

and one measurement (site A) and two measurements (site B) on PM₁₀. Regarding mold, one mold count on the dust of ventilation ducts was carried out and airborne mold concentrations were measured six times in each of the three rooms investigated of site B and outdoor, and two times in each of the two rooms of site A and outdoor.

43 VOCs, 46 SVOCs, and 4 genera of molds were analyzed. List of pollutants is presented in Table 2. CAS numbers of VOCs and SVOCs are available in Tables S1-S2. Measurements focused on VOCs commonly studied in indoor environments, measured by OQAI, or VOCs from mandatory labeling on construction products, according to their toxicity. Selection of SVOCs was based on a health-based ranking of SVOCs established according to contamination data and reference doses,⁴³ on published concentrations in indoor environment and on the technical feasibility. Regarding mold measurement, the

genera most frequently found in the air and dust of dwellings were studied.⁴⁴

3 | SAMPLING METHOD

3.1 | Ambient parameters

Temperature and relative humidity were measured continuously in each room with recorders HOBOS (Onset), outdoor under a box with three faces behind the construction panel and indoor, attached by a scratch system on the glazing (A0, A1, B0, B1) or attached on a tripod in the center of the room (B2). The time interval of the measurements was 15 minutes.

TABLE 2 List of pollutants analyzed

Types of pollutants (number of substances)		Substances
Chemical		
VOCs (43)	Aldehydes (11)	- Formaldehyde, acetaldehyde, propionaldehyde, butyraldehyde, benzaldehyde, isovaleraldehyde, valeraldehyde, hexaldehyde, o-tolualdehyde, m-tolualdehyde, p-tolualdehyde
	Alcohols (4)	- Ethanol, isopropanol, n-propanol, 2-ethyl-1-hexanol
	Ketones (2)	- Acetone, 2-butanone
	Chlorobenzenes (1)	- 1,4-Dichlorobenzene
	Ethers (1)	- Ether
	Glycol ethers (3)	- 2-butoxyethanol, 2-ethoxyethanol, phenoxyethanol
	Aliphatic hydrocarbons (3)	- n-heptane, n-decane, n-undecane
	Halogenated hydrocarbons (7)	- Chloroform, 1,1,1-trichloroethane, trichloroethylene, bromodichloromethane, tetrachloroethylene, dibromochloromethane, tribromomethane
	Aromatic hydrocarbons (9)	- Benzene, toluene, ethylbenzene, xylenes (o, m, p), styrene, 1,2,4-trimethylbenzene, naphthalene
	Phenols (1)	- Phenol
	Terpenes (1)	- Limonene
SVOCs (46)	Phthalates (6)	- Benzyl butyl phthalate (BBP), dibutyl phthalate (DBP), di(2-ethylhexyl)phthalate (DEHP), diethyl phthalate (DEP), diisononyl phthalate (DiNP), diisobutyl phthalate (DiBP)
	Musks (2)	- Galaxolide, tonalide
	Pyrethroids (1)	- Permethrin
	Organochlorine pesticides (7)	- Aldrin, dieldrin, 4,4'-DDT, 4,4'-DDE, alpha-ICH, lindane, alpha-endosulfan
	Organophosphorous pesticides (2)	- Ethyl chlorpyrifos, diazinon
	Others pesticides (1)	- Oxadiazon
	Phosphoric esters (1)	- Tributyl phosphate
	Polycyclic aromatic hydrocarbons (PAHs) (11)	- Anthracene, benzo[a]pyrene, fluoranthene, fluorene, phenanthrene, pyrene, benzo[b]fluoranthene, benzo[k]fluoranthene, dibenzo[a,h]anthracene, indeno[1,2,3-cd]pyrene, benzo[a]anthracene
	Polychlorobiphenyls (PCBs) (9)	- PCB 28, PCB 31, PCB 52, PCB 101, PCB 105, PCB 118, PCB 138, PCB 153, PCB 180
Microbiological	Polybromodiphenylethers (PBDEs) (5)	- PBDE 47, PBDE 85, PBDE 99, PBDE 100, PBDE 153
	Others (1)	- Triclosan
Mold (4)		
		- <i>Aspergillus</i> , <i>Penicillium</i> , <i>Cladosporium</i> , <i>Alternaria</i>

3.2 | TVOCs

Total volatile organic compounds (TVOCs) were measured continuously by a Fireflies station (Hager Services), attached on a tripod. This station measured the overall compound content with a carbon chain from C3 to C10. Results are expressed as toluene equivalent and taken into account the following gases: nitric oxide, nitrogen dioxide, sulfur dioxide, ammonia, and carbon monoxide. The station also measured temperature and relative humidity.⁴⁵ The time interval of the measurements was 10 minutes.

3.3 | Aldehydes

During the two first construction stages on site A, aldehydes were collected during 4 days using commercially available Radiello passive samplers (RAD165, Sigma-Aldrich, Saint-Louis) impregnated with 2,4-dinitrophenylhydrazine (DNPH). The other measurements were active sampling, on Waters Sep-Pak XPOsure cartridges (WAT047205, Waters) with silica impregnated with DNPH. Air was pumped (Gilliam 3500; Sensidyne) at a constant flow rate of 500 mL/min during 390 minutes, and the pump was attached on a tripod. Samples were stored at 5°C until analysis.

3.4 | Other VOCs

During the two first construction stages on site A, other VOCs were collected during 4 days using commercially available Radiello passive samplers (RAD145; Sigma-Aldrich) filled with graphitized charcoal (Carbograph 4). The other measurements were active sampling on tubes filled with adsorbant Carboxen B (20273; Sigma-Aldrich) and Carboxen 1000 (10478-U; Sigma-Aldrich) with a pump (Gilliam LFS-113; Sensidyne) attached on a tripod at a constant flow rate of 50 mL/min during 180 minutes. Samples were stored at 5°C until analysis.

3.5 | PM₁₀

Airborne PM₁₀ particles were pumped (GilAir-5 Pump; Sensidyne) at a constant flow rate of 3L/min through a MicroVol device (Ecomasure) on Teflon filters (47 mm diameter and 2 µm porosity, Pall Teflon) during 72 hours on site A and through an impactor on 37-mm quartz fiber filters (TERA Environnement, France) during 96 hours on site B. Samples were stored at -18°C until SVOC analysis.

3.6 | Dust from ventilation ducts

Dust from ventilation ducts was collected with wipes (ASTM E1792 Lead Wipes; Aramsco) for the analysis of SVOCs and swabs

(020063; Dutscher) for the analysis of mold. Before sampling, 1 mL of saline solution of Tween® 80 (Merck KGaA) was added to swabs.

3.7 | Mold

Molds were collected by bioimpaction (Sampl'air biocollector; AES Laboratory). 50 liters of air was impacted on two culture media: Malt Extract Agar (Biokar Diagnostics, France) and Dichloran Glycerol 18% Agar (Merck KGaA, Darmstadt) (sampling time: 30 seconds).

4 | ANALYTICAL METHOD

4.1 | Aldehydes

Aldehydes were extracted with acetonitrile (2 mL for Radiello samplers and 5 mL for Waters Sep-Pak XPOsure cartridges) and analyzed by high-performance liquid chromatography (1260 Infinity Liquid Chromatography System, Agilent Technologies) coupled to UV detection (1100 Series Diode Array Detector, Agilent Technologies). The analytical method is described by Baurès et al.⁴⁶ Limits of quantification (LOQ) are given in Table S3.

4.2 | Other VOCs

The other VOCs were thermally desorbed (Perkin Elmer ATD Turbo Matrix 150) and analyzed by gas chromatography (7890A, Agilent Technologies) coupled to mass spectrometry (Agilent Technologies 5975C inert XI EI/CI MSD). The analytical method is described by Baurès et al.⁴⁶ LOQ are given in Table S4.

4.3 | SVOCs

SVOCs were desorbed from PM₁₀ and dust by thermal extraction (GERSTEL Thermal Desorption Unit TDU 2, GERSTEL GmbH & Co. KG) and analyzed by gas chromatography (7890A, Agilent Technologies) coupled to tandem mass spectrometry (70000B Triple Quad, Agilent Technologies). The analytical method is described by Blanchard et al and Mandin et al.^{16,47}

4.4 | Mold

The cultures were incubated (Fricell, MMM Group) for 5 to 7 days at room temperature (25°C ± 2°C), and the fungal thalli were identified from their characteristics observed under optical microscopy (CH-2, Olympus) and quantified.

TABLE 3 Volatile organic compound concentrations ($\mu\text{g}/\text{m}^3$) in air

Concentrations (µg/m ³) ^a															
VOCs	Initial state					Preparation of supports					Finish paints				
	A0	A1	B0	B1	B2	A0	A1	B0	B1	B2	A0	A1	B0	B1	B2
Aldehydes															
Formaldehyde	—	—	—	—	—	5.4	6.2	—	5.5	2.4	21.8	70.8	15.5	6.1	13.5
Acetaldehyde	2.2	—	—	—	—	8.1	9.2	7.5	4.8	2.9	16.4	43.4	140.2	22.4	300.2
Propionaldehyde	—	—	—	—	15.7	—	2.7	—	—	—	—	22.2	2.3	—	3.2
Butyraldehyde	10.3	10	—	—	—	—	—	3	2.1	—	—	12	2.5	—	6
Benzaldehyde	—	1.1	—	68.3	21.7	1.6	1.3	—	—	—	3.4	6	6.4	7.3	28.9
Isovaleraldehyde	—	—	—	—	—	2	2.3	—	—	—	—	6.3	3.9	—	6
Valeraldehyde	—	—	—	—	—	—	—	—	—	—	4.5	63.4	7.3	—	—
Hexaldehyde	—	—	—	—	—	16.3	18	9.3	6.3	5.2	22.2	497.3	7.9	3.6	9.7
o-Tolualdehyde	NT	NT	—	—	—	NT	NT	—	—	—	9.1	—	—	—	—
m-Tolualdehyde	NT	NT	—	—	—	NT	NT	2.5	—	—	—	—	—	—	13.2
Alcohols															
Ethanol	NT	NT	—	—	—	NT	NT	—	—	—	61.98 ^b	190.73 _b	—	—	5.47
Isopropanol	NT	NT	—	—	—	NT	NT	—	—	—	1.72	7.3	—	—	—
2-Ethyl-1-hexanol	NT	NT	—	—	—	NT	NT	—	—	—	—	—	—	—	—
Ketones															
Acetone	NT	NT	—	—	—	NT	NT	—	—	2.48	—	1.33	—	—	—
2-butanone	NT	NT	10.34	16.5	5.81	NT	NT	—	—	—	5.28	22.93	—	—	—
Ethers															
Ether	NT	NT	—	—	—	NT	NT	—	—	—	—	—	—	—	5.65
Glycol ethers															
2-Butoxyethanol	NT	NT	—	—	—	NT	NT	—	—	—	—	—	—	—	—
Phenoxyethanol	NT	NT	—	—	—	NT	NT	—	—	—	—	—	—	5.31	—
Aliphatic hydrocarbons															
n-Heptane	NT	NT	—	—	—	NT	NT	—	—	—	—	—	—	—	—
n-Decane	NT	NT	—	1.64	4.01	NT	NT	—	—	—	—	0.54	0.52	—	0.59
n-Undecane	NT	NT	—	0.71	0.81	NT	NT	—	—	—	—	0.96	0.48	0.21	0.66
Halogenated hydrocarbons															
1.1.1. -Trichloroethane	NT	NT	—	—	—	NT	NT	—	—	—	—	—	—	—	—
(Continues)															

(Continues)

TABLE 3 (Continued)

Concentrations (µg/m ³) ^a															
VOCs	Initial state			Preparation of supports						Finish paints					
	A0	A1	B0	B1	B2	A0	A1	B0	B1	B2	A0	A1	B0	B1	B2
1.1.2.2-Tetrachloroethylene	—	0.2	—	—	—	—	—	—	—	—	—	—	—	—	—
Aromatic hydrocarbons															
Benzene	1.48	1.56	—	—	—	1.69	1.71	—	—	—	—	—	—	—	—
Toluene	4.6	5.73	—	0.54	0.61	4.37	5.07	—	—	—	—	—	0.37	—	0.27
Ethylbenzene	2.61	3.11	1.62	10.75	6.26	1.81	1.93	0.51	1.01	0.26	—	0.94	1.8	0.96	3.87
m,p-Xylene	9.71	10.27	4.67	43.48	23.44	5.14	5.48	1.74	3.77	0.66	0.64	2.26	4.03	1.89	4.97
o-Xylene	3.48	3.85	1.74	17.1	10.31	2.7	2.8	0.88	1.68	0.32	0.37	1.05	1.97	0.84	2.51
Styrene	NT	NT	—	0.88	1.09	NT	NT	0.26	—	0.22	1.61	21.28	0.58	0.23	1.29
1,2,4-Trimethylbenzene	NT	NT	—	0.83	2.22	NT	NT	—	—	—	—	—	—	—	—
Naphthalene	NT	NT	—	—	—	NT	NT	—	—	—	—	0.21	—	0.21	—
Phenols															
Phenol	NT	NT	—	—	—	NT	NT	—	—	—	—	—	—	0.53	—
Terpenes															
Limonene	NT	NT	—	2.69	3.18	NT	NT	0.39	0.46	—	—	0.33	0.55	—	0.5
TVOCs identified	34.38	35.82	18.37	163.42	95.14	49.11	56.69	26.08	25.62	14.44	87.02	780.53	196.3	49.58	406.48
Concentrations (µg/m ³) ^a															
Soft floors															
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TABLE 3 (Continued)

Concentrations (µg/m ³) ^a															
Soft floors				Last finishing works								Delivery			
				B0 (living room)											
VOCs	A0	A1	B0	B2	A0	A1	B0	B1	B2	A0	A1	B0	B1	B2	B2
o-Tolualdehyde	—	14.2	—	—	—	—	—	2.3	5.1	—	—	—	—	—	—
m-Tolualdehyde	—	—	3.9	3.2	—	—	2.4	—	—	—	—	—	—	—	—
Alcohols															
Ethanol	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Isopropanol	—	—	—	—	—	—	9.5	—	9.3	—	—	—	—	—	—
2-Ethyl-1-hexanol	—	—	—	—	5.67	—	—	—	16.2	—	—	—	—	—	7.2
Ketones															
Acetone	—	—	—	5.44	—	—	1.4	1.6	15.1	—	—	—	—	—	—
2-butanone	8.73	4.63	—	—	—	—	—	—	20.1	—	—	—	—	—	—
Ethers															
Ether	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Glycol ethers															
2-Butoxyethanol	—	—	—	—	—	—	—	—	—	6.25	5.88	—	—	—	—
Phenoxyethanol	—	10.53	—	—	—	6.31	—	—	6.1	>101.01	26.81	—	—	—	—
Aliphatic hydrocarbons															
n-Heptane	—	—	—	—	—	—	—	—	1.4	—	—	—	—	—	—
n-Decane	20.3	33.06	—	0.33	1.31	1.28	14.2	0.2	>22	—	—	0.9	0.2	8.2	—
n-Undecane	20.9	27.19	—	0.56	1.44	1.37	6.3	—	>22	0.29	—	1.1	0.2	9.9	—
Halogenated hydrocarbons															
1,1,1-Trichloroethane	—	—	—	0.6	—	—	—	—	—	—	—	—	—	—	—
1,1,2,2-Tetrachloroethylene	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Aromatic hydrocarbons															
Benzene	—	—	—	—	—	—	—	—	24.17	—	—	—	—	—	—
Toluene	0.62	—	—	2.11	0.28	0.27	2.4	—	11.2	—	0.98	—	—	—	1.8
Ethylbenzene	4.13	2.21	—	2.11	0.56	0.25	2.1	—	10.6	—	—	—	—	—	1.3
m,p-Xylene	14.08	8.45	—	0.78	1.78	0.93	8.4	—	40.7	—	—	0.8	—	—	5.9
o-Xylene	6.18	4.83	—	0.44	0.73	0.44	4.9	—	18	—	—	0.6	—	—	3.4
Styrene	0.42	33.06	—	1	0.26	—	0.4	—	2.9	—	—	15.1	13.5	>22	>22

(Continues)

TABLE 3 (Continued)

Concentrations (µg/m ³) ^a														
	Soft floors				Last finishing works					Delivery				
	B0 (living room)													
VOCs	A0	A1	B0	B2	A0	A1	B0	B1	B2	A0	A1	B0	B1	B2
1,2,4-Trimethylbenzene	7.95	11.11	—	—	0.83	0.65	3	—	9	—	—	0.3	—	2.5
Naphthalene	—	—	—	—	—	—	—	—	0.2	—	—	—	—	—
Phenols														
Phenol	—	—	—	—	—	—	—	—	2.1	5.34	6.25	—	—	0.7
Terpenes														
Limonene	—	—	—	1.11	—	0.41	0.8	—	2.1	>20.20	3.52	0.6	—	4.8
TVOCs identified	258.01	327.37	98.6	85.72	114.66	83.01	115.1	74.7	294.57	46.18	101.64	39	102.8	128.4

Abbreviation: VOCs: volatile organic compounds.

^aA0: ground floor site A, A1: first floor site A, B0: ground floor site B, B1: first floor site B, B2: second floor site B.^bImpact of alcohol used for disinfection for mold sampling; NT: non tested; "—": value below limit of quantification.

5 | RESULTS

5.1 | VOCs

The VOC concentrations at each construction stage are presented in Table 3. Nine compounds are always under the LOQ, p-tolualdehyde, n-propanol, 1,4-dichlorobenzene, 2-ethoxyethanol, chloroform, bromodichloromethane, dibromochloromethane, tribromomethane, and trichloroethylene and are not presented. The following compounds have between 50% and 70% of their samples above the LOQ: propionaldehyde, butyraldehyde, valeraldehyde, n-decane and n-undecane, toluene, styrene, and limonene. The following compounds have more than 70% of their samples above the LOQ: formaldehyde, acetaldehyde, benzaldehyde, hexaldehyde, ethylbenzene, and xylenes.

Finish paints, soft floors, and last finishing works were the stages with the highest variety of VOCs identified. The application of finish paints was characterized by the highest concentrations of aldehydes. Aromatic hydrocarbons were found during all the construction process. Aliphatic hydrocarbons (n-decane, n-undecane) were present during the application of soft floors on site A, and these compounds were also found on B2 during the last finishing works. At delivery, glycols ethers and terpenes were identified on site A.

Overall, VOC concentrations were higher on site A, especially on A1. This was confirmed by the results of TVOC's measurements (Table 4). The implementation of soft floors seemed to be the stage with the highest TVOC levels. TVOC concentrations measured were well above the TVOCs' identified levels.

Figure 1 presents the VOC proportions at each construction stage according to the chemical classes. They were calculated by the sum of VOCs in each chemical class divided by the sum of all VOCs measured on the same site/floor. Chemical classes of alcohols, ketones, ethers, halogenated hydrocarbons, phenols, and terpenes are grouped in the category "others" because of the very low concentrations of these compounds. The average concentration at the three floors (ground floor, first floor, and second floor) was used for site B, while the distinction between the garden ground and the first floor was retained for Site A because of a shift in the progress of construction works. This figure allows to identify the majority chemical classes at each construction phase on each site.

Aromatic hydrocarbons and aldehydes were the two main chemical classes of VOCs at the two first construction stages. Aldehydes were the main chemical classes in almost all the other construction stages. During the implementation of soft floors, there were a large amount of aliphatic hydrocarbons. During the last finishing works, aliphatic hydrocarbons were in high concentrations on site B. At delivery, there was an important percentage of glycol ethers on site A.

The evolution of the concentrations of the two main aldehydes measured at delivery is presented in Figures 2 and 3.

On site A, a peak of formaldehyde was observed during the application of finish paint, especially on A1. A second peak was observed on A0 and A1 during the last finishing works. Formaldehyde concentrations remained relatively high at delivery despite the

start of the ventilation system, contrary to the site B where the ventilation system was not started. On Site B, formaldehyde concentrations started to increase during the application of finish paints and the peak was observed during the implementation of soft floors.

A peak of hexaldehyde was observed on A1 during the application of finish paints. A shift in the application of finish paints was observed between A0 and A1, and it explains that a peak was observed on A0 during the implementation of soft floors. On site B, levels increased with the implementation of soft floors and the last finishing works to stay relatively high at delivery.

5.2 | SVOCS

Results of SVOC measurements are presented in Table 5. Results with the LOQ of our samples are presented in Table S5. 22 compounds were never quantified above the LOQ: aldrin, dieldrin, 4,4'-DDT, 4-4'-DDE, lindane, alpha-endosulfan, ethyl chlorpyrifos, diazinon, PCBs 28, 31, 52, 101, 105, 118, 138, 153, and 180, and PBDEs 47, 85, 99, 100, and 153.

Three polycyclic aromatic compounds (PAHs) were found in the dust from ventilation ducts: fluoranthene, fluorene, and phenanthrene. Other PAHs were found on airborne PM₁₀. At each measured stage, phthalates were found in high concentrations on PM₁₀. DiNP and DEHP were predominant. Permethrin was also present. Musks were found on site B, especially at delivery. Triclosan was detected at each measured stage.

5.3 | Mold

Results of mold analysis and associated ambient parameters measured with HOBOS are presented in Table 6.

The first two stages were characterized by a high humidity rate, reaching 80% on the two sites. During the preparation of supports, concentrations of mold indoor were higher than outdoor, especially on site A with a concentration reaching 1400 CFU/m³ in A1 when the median percentage of humidity was the highest (81%). On site B, all genera (*Aspergillus*, *Penicillium*, *Cladosporium*, and *Alternaria*) were in higher concentrations indoor than outdoor. During the application of finish paints, the airborne mold concentrations were twofold higher on A0, reaching 1000 CFU/m³. At delivery on site B, the mold concentrations indoor were lower than outdoors. All along the construction process, *Cladosporium* was the most commonly detected genus.

No mold was detected in ventilation ducts during the application of finish paints on ground floor on site A, but the presence of mold was revealed on the first floor (1 CFU/dm²).

6 | DISCUSSION

Overall, concentrations were higher on site A because this construction site was more confined than site B and succession of construction stages was quicker. The comparison between the two sites highlights the importance of natural ventilation, a delay between the sequence of tasks, the materials, and product selection used during

TABLE 4 Total volatile organic compounds, temperature, and relative humidity measured with Fireflies station in air during the VOC measurement campaigns

Construction stage	Construction site	Measurement campaigns	TVOC (µg/m ³)			Temperature (°C)			Hygrometry (%)		
			Median	Min	Max	Median	Min	Max	Median	Min	Max
Initial state	A	February 22 to 26, 2016	—	—	—	—	—	—	—	—	—
	B	January 10, 2017	2098	1678	3187	7.2	6.8	7.8	86	83	89
Preparation of supports	A	April 18 to 22, 2016	6805	2658	15 350	15.8	14.5	17.5	84	73	86
	B	April 13, 2017	1004	940	1170	16.7	16.5	17.3	58	56	59
Finish paints	A	May 13, 2016	13 410	5227	14 630	19	18.7	19.3	84	81	85
		May 25, 2016	19 070	13 140	38 250	19.7	19.5	20.2	69.5	66	77
	B	May 2, 2017	2741	2004	3542	14.5	14.2	15.2	78	75	80
Soft floors	A	June 3, 2016	52 780	51 250	53 880	19.8	19.3	19.9	85	85	86
		June 8, 2016	23 310	17 940	30 460	19.8	19.3	19.9	85	85	86
	B	August 4, 2017	1642	1451	2054	24.1	23.9	24.6	66	65	70
Last finishing works	A	June 21 to 24, 2016	5745	4471	7745	20.7	20.2	21.2	84	83	85
	B	September 19, 2017	—	—	—	—	—	—	—	—	—
Delivery	A	July 27, 2016	—	—	—	—	—	—	—	—	—
	B	October 3, 2017	—	—	—	—	—	—	—	—	—

Abbreviation: "—": data not available.

the construction process to reduce the concentrations of pollutants in indoor air.

6.1 | VOCs

Concentrations of VOCs measured on site A at delivery were compared to the medians of concentrations measured in office buildings included in the European project OFFICAIR.⁴⁸ This European project comprises indoor air quality measurements in office buildings distributed among eight participated countries. 37 office buildings were included in the summer campaign (2012) and 35 in the winter campaign (2012-2013), a total of 148 rooms were investigated.

Concentrations of VOCs on site B at delivery were compared to the medians of the annual average concentrations measured in energy-efficient buildings from the OQAI's national indoor air quality reference database.⁴⁹ The study was carried out on the basis of validated and complete survey data included in the database on May 2015. 72 dwellings were investigated and they had an average air permeability of $0.54 \text{ m}^3/(\text{h} \cdot \text{m}^2)$ for individual houses and 0.86 for collective dwellings to 4 Pa , approximatively as the buildings investigated in this project.

The comparison between the concentrations measured in this project and these two other databases is presented in Table 7.

For site A, data are of the same order of magnitude between both projects except concentrations of formaldehyde and 2-butoxyethanol which reached twice the values observed in the project OFFICAIR.

For site B, most of the compounds were measured at concentrations of the same order of magnitude of the OQAI's values. Concentrations are four times lower than the OQAI's data for

formaldehyde, and concentrations of hexaldehyde reached twice the value of OQAI. Concentrations of styrene are much higher than the OQAI's value. Finally, concentrations of limonene are well under the OQAI's value.

The low concentrations of benzene and trichloroethylene may be due to the regulation of 2009 on the conditions for placing on the market construction and decoration products containing carcinogenic, mutagenic, and reprotoxic substances.⁴⁰

In this project, the overall compound contents measured with the Fireflies station are well above the sum of each VOCs' identified concentration. It suggests that either gas detection interferes with the measurement or a significant portion of VOCs was not identified in this project. Therefore, for future investigations, it would be necessary to expand measurements to other volatile compounds. Moreover, sensor sensitivity to alcohols (used for disinfection during mold sampling) has been highlighted during the project.

Regarding potential sources of formaldehyde, wood-based panels are an important source of formaldehyde.⁵⁰ Melamine wood panels have been installed in the sill of all the windows on site A, which could explain the difference in formaldehyde concentrations between the two sites. About hexaldehyde, measured levels at delivery on site B could be linked to the wooden frame façade. Considering their high mobility, VOCs could penetrate through materials and diffuse into buildings. High concentrations of hexaldehyde with wood frame façade have also been observed during air quality measurements at delivery of an eco-district in Langoüet, France.⁵¹ During the measurement campaign carried out during the installation of soft floors, the storage of soft floors in the living room on B0 (room used for the storage of soft floors before their installation) does not seem to have had an impact on the concentrations of VOCs in the air, as the concentrations were not different from the measurements in the

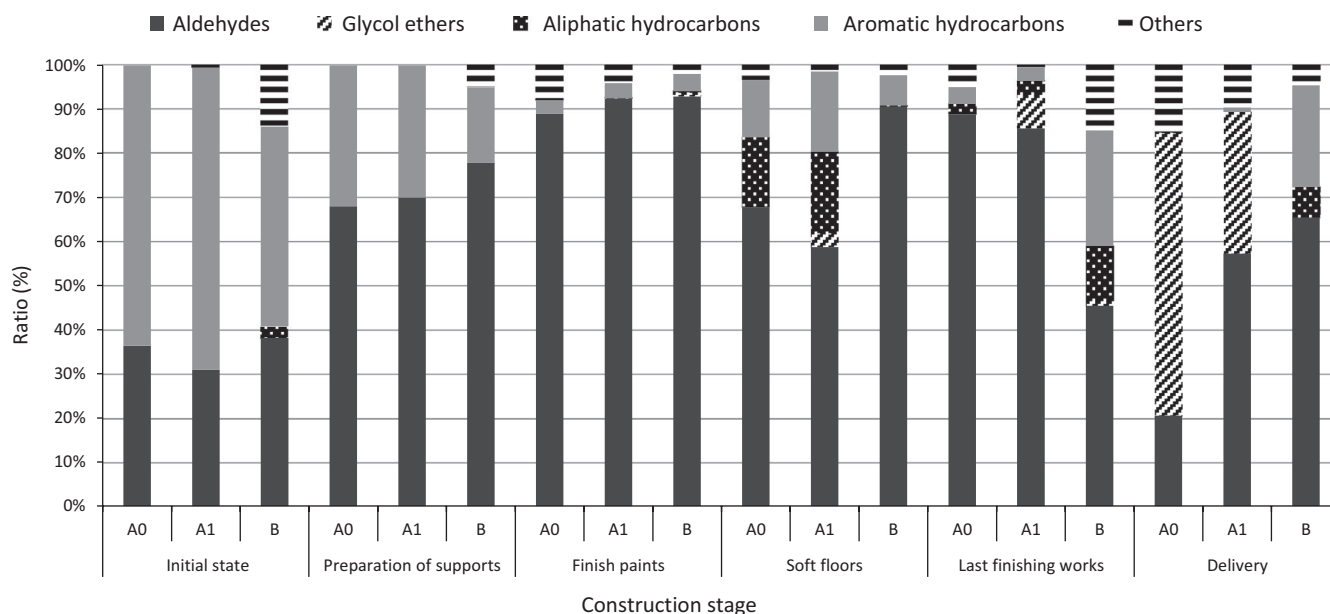


FIGURE 1 Proportions of volatile organic compound chemical classes in air at each construction stage on site A (ground floor, first floor) and site B

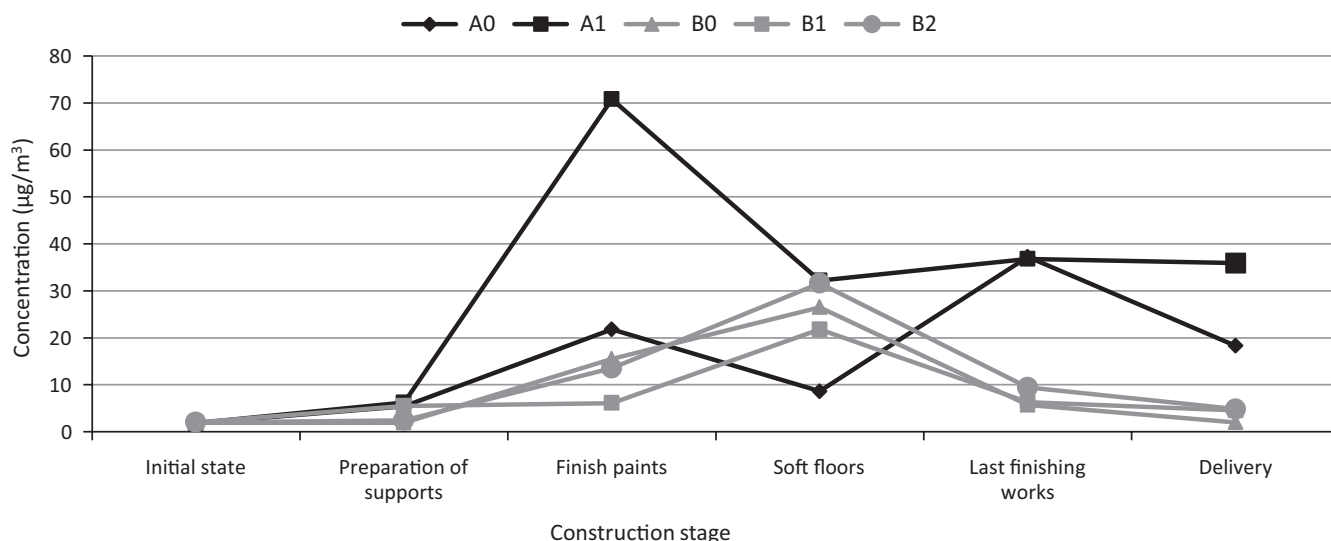


FIGURE 2 Formaldehyde concentrations in air at each construction stage on site A (ground floor, first floor) and site B (ground floor, first floor, second floor)

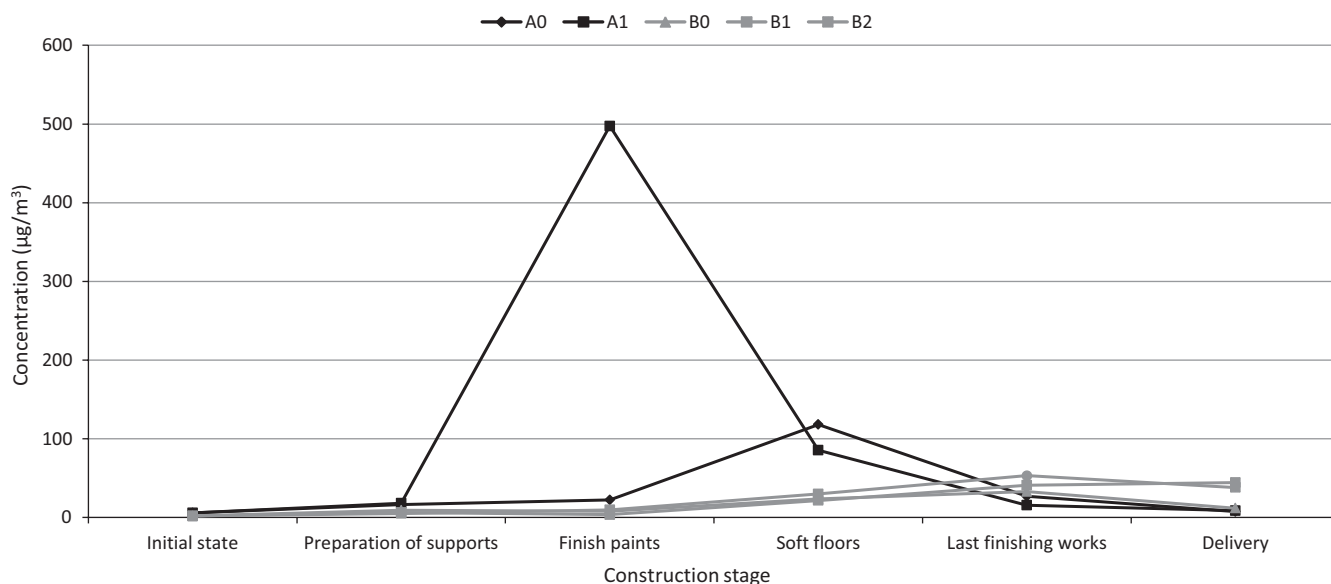


FIGURE 3 Hexaldehyde concentrations in air at each construction stage on site A (ground floor, first floor) and site B (ground floor, first floor, second floor)

other rooms. Glycol ethers and limonene present at delivery on site A were probably emitted by the cleaning products.^{52,53} Professional cleaning products were used on site A before delivery for dust removal, removal of stains (cement), labels on certain products, cleaning of floors and glazing and joinery, and disinfection of toilets (mainly on site A as it is an establishment open to the public). On site B, less-emissive processes were used, including steam cleaning.

In order to assess the potential health impact for future occupants related to the presence and levels of the studied pollutants in indoor air, a comparison with recent guideline values applicable in occupied indoor environments was carried out. Values are reported in Table S6. Some IAGV, strictly based on health criteria, were proposed by the European

Commission or by the French Agency for Food, Environmental and Occupational Health & Safety (Anses).⁵⁴⁻⁵⁹ In 2009, the French High Council for Public Health (HCSP) recommended reference values for management support in confined spaces for new buildings delivered from 2012, as well as to initiate corrective actions if necessary in existing buildings.^{60,61} In France, regulatory values have been set for benzene and formaldehyde and must not exceed indoors.⁶²

At delivery, concentrations measured are well under the guideline values for acetaldehyde, benzene, ethylbenzene, styrene, toluene, and xylenes. The IAGV short term of 100 µg/m³ proposed by Anses for formaldehyde has not been exceeded, but the concentration on A1 reached the regulatory value and the reference value of

SVOCs	Dust ^a (ng/g)	Airborne PM ₁₀ (pg/m ³) ^b		
	A	A1	B2	
	Finish paints	Last finishing works	Soft floors	Delivery
Phthalates				
BBP	—	—	379.8	495.4
DBP	—	426	740.9	11 740
DEHP	—	10,440	8325	26 210
DEP	—	—	2923	7352
DiNP	—	>38 580	8798	23 120
DiBP	—	185	672.7	11 190
Musks				
Galaxolide	—	—	43.2	127.1
Tonalide	—	—	8.5	29
Pyrethroids				
Permethrin	—	77	16.9	41.4
Organochlorine pesticides				
Alpha-HCH	—	—	5.2	—
Others pesticides				
Oxadiazon	—	—	—	1.4
Phosphoric esters				
Tributyl phosphate	—	39	—	174.1
Polycyclic aromatic hydrocarbons (PAHs)				
Anthracene	—	13	7	19.3
Benzo[a]pyrene	—	25	5.1	101.9
Fluoranthene	50	52	20.6	38.2
Fluorene	95	19	26.4	50.8
Phénanthrene	303	64	37.3	50.1
Pyrène	—	35	14.4	46.4
Benzo[b]fluoranthene	—	46	14.9	128
Benzo[k]fluoranthene	—	9	—	22
Dibenzo[a,h]anthracene	—	—	—	14.7
Indeno[1.2.3-cd]pyrene	—	21	8.6	142.8
Benzo[a]anthracene	—	13	—	36.3
Others				
Triclosan	—	38	6.5	11.1

Abbreviations: BBP: benzyl butyl phthalate; DBP: dibutyl phthalate; DEHP: bis(2-ethylhexyl) phthalate; DEP: diethyl phthalate; DiBP: diisobutyl phthalate; DiNP: diisononyl phthalate; PAHs: polycyclic aromatic hydrocarbons; SVOCs: semi-volatile organic compounds; α -HCH: α -hexachlorocyclohexane.

^afrom ventilation ducts.

^bA1: ground floor site A, B2: second floor site B; "—": value below limit of quantification.

TABLE 5 Semi-volatile organic compound concentrations adsorbed on dust from ventilation ducts and adsorbed on airborne particles

the HCSP of 30 $\mu\text{g}/\text{m}^3$.^{61,62} The value of 10 $\mu\text{g}/\text{m}^3$ proposed as target value to be achieved in 10 years by the HCSP was exceeded on A0 and A1. In France, this value will be a regulatory value in 2023.

On sites A and B, with the evolution of construction materials (use of less-emissive materials) and with the application of the 2012

French Thermal Regulation and considering the VOCs individually, the measured levels respected the guide values at the delivery of buildings, but not the future regulatory value applicable in 2023. Nevertheless, concentrations measured at delivery (therefore over a short period of time) are compared with guideline values intended

TABLE 6 Mold count on dust and airborne mold concentrations with temperature and relative humidity associated

Construction stages	Location ^a	Mold count on dusts ^b (CFU/dm ²)	Airborne mold concentrations (CFU/m ³)					Temperature (°C)			Hygrometry (%)		
			Mold	Aspergillus	Penicillium	Cladosporium	Alternaria	Median	Min	Max	Median	Min	Max
Initial state	B outdoor	—	350	100	<20	200	<20	8.37	7.52	11.54	75.85	62.68	82.9
	B0	—	510	<20	360	150	<20	7.34	6.91	10.96	76.77	62.22	78.85
	B1	—	230	20	80	110	20	—	—	—	—	—	—
	B2	—	240	70	<20	110	<20	7.59	6.86	11.59	77.78	58	82.85
Preparation of supports	A outdoor	—	340	—	—	—	—	11.61	7.59	20.82	72.63	45.57	85.36
	A0	—	860	—	—	—	—	13.22	11.35	26.18	73.72	44.59	77.86
	A1	—	1400	—	—	—	—	13.56	12.2	15.99	81.28	75.82	84.36
	B outdoor Feb 28	—	250	20	50	30	<20	7.67	3.14	11.66	78.26	56.94	88.37
	B0 Feb 28	—	290	20	170	40	20	9.78	8.99	10.91	81.27	78.32	85.08
	B1 Feb 28	—	670	50	120	390	30	—	—	—	—	—	—
	B2 Feb 28	—	480	50	80	150	30	9.39	8.47	9.93	69.47	63.74	73.41
	B outdoor Apr 14	—	510	20	110	320	20	11.69	5.67	16.51	68.96	50.96	82.6
	B0 Apr 14	—	180	<20	40	100	<20	14.76	12.92	18.94	59.00	52.74	68.94
	B1 Apr 14	—	340	60	<20	220	<20	13.86	11.86	14.53	62.04	58.27	65.52
Finish paints	B2 Apr 14	—	460	70	50	230	<20	16.19	15.75	18.3	58.77	55.62	62.66
	A ground floor	Absence											
	A first floor 1												
	A outdoor	—	880	—	—	—	—	14.18	12.92	16.56	91.25	87.77	91.61
Soft floors	A0	—	1000	—	—	—	—	17.65	17.15	18.91	75.86	72.31	77.32
	A1	—	520	—	—	—	—	18.43	18.03	19.18	81.06	79.21	82.07
	B outdoor	—	110	1	2	82	<1	20.15	16.75	22.82	81.26	68.17	87.18
	B0	—	26	3	4	16	<1	22.73	22.27	23.64	69.33	66.94	74.74
	B0 (living room)	—	31	1	3	21	<1	—	—	—	—	—	—
	B2	—	22	2	3	17	<1	24.17	23.88	24.75	64.36	61.57	67.83
	B outdoor	—	990	<20	110	570	30	—	—	—	—	—	—
Last finishing works	B0	—	500	20	290	150	20	—	—	—	—	—	—
	B1	—	390	<20	50	270	<20	—	—	—	—	—	—
	B2	—	290	20	50	160	<20	—	—	—	—	—	—

(Continues)

TABLE 6 (Continued)

Construction stages	Location ^a	Mold count on dusts ^b (CFU/dm ²)	Airborne mold concentrations (CFU/m ³)					Temperature (°C)			Hygrometry (%)		
			Mold	Aspergillus	Penicillium	Cladosporium	Alternaria	Median	Min	Max	Median	Min	Max
Delivery	B outdoor	—	>2000	>2000	>2000	>2000	>2000	—	—	—	—	—	—
	B0	—	580	<20	60	420	<20	—	—	—	—	—	—
	B1	—	580	<20	60	320	<20	—	—	—	—	—	—
	B2	—	380	20	60	300	<20	—	—	—	—	—	—

^aAO: ground floor site A, A1: first floor site A, B0: ground floor site B, B1: first floor site B, B2: second floor site B.

^bfrom ventilation ducts; "—": data not available; CFU: colony-forming units.

for long-term exposure, such as for formaldehyde and benzene. This is a limitation that is important to note.

6.2 | SVOCs

Regarding dust from ventilation ducts, dust can accumulate on ducts not only during the manufacturing process but also during the various phases of storage, transportation, and construction.⁶³ In this project, PAHs measured in the dust from ventilation ducts probably came from degreasing oils and solvents used in the factory for the manufacture of ventilation ducts, as described by Holopainen et al and Asikainen et al.^{64,65} This project supports the recommendation that ducts should be cleaned and protected against impurities during transport, storage, and construction.^{64,66,67}

Regarding PM₁₀, the high concentrations of PAHs measured at delivery on site B can be explained by the use of tobacco reported on site B.^{68,69} Regarding the high diversity of phthalates, the dominant plasticizer for flexible polyvinyl chloride flooring was DEHP for many years, but DiNP replaced phthalates such as DEHP, DBP, and BBP, due to their potential adverse health effects.^{15,16} A restriction of their use was observed in several products in Europe, and an emission threshold of 1 µg/m³ has been implemented for DEHP and DBP. About pesticides, permethrin is known to be used for wood protection.⁷⁰ About antimicrobial agent, triclosan is widely used in various personal care and consumer products.^{71,72} No data are available on triclosan concentrations during the construction process, but Mandin et al measured SVOCs on PM10 filters collected over 7 days during a nationwide survey of 285 French dwellings and found that triclosan was one of the major particle-bound SVOCs with a median concentration of 114 pg/m³.⁴⁷ Measurements were conducted from October 2003 to December 2005. In 2014, it has been banned for use in the treatment of construction products and materials since a decision made by the European Commission.⁷³ Triclosan is an endocrine disruptor and is associated with reproductive and developmental outcomes in animal and in vitro studies, and toxicological studies suggest that it can affect endocrine function and antibiotic resistance in human.⁷⁴ In this study, triclosan was found at the three construction stages measured. The source of triclosan has not been identified in this project.

6.3 | Mold

The first stages after the structural work seem to be critical stages in relation to the increase in moisture content. This is related to the increase in mold concentrations which reached and exceeded the value of 1000 CFU/m³ proposed by Anses as abnormal concentration in total molds and requiring professional intervention for remediation.⁷⁵

These observations raise questions about the management of moisture on construction sites as long as exposure to mold in a work

TABLE 7 Comparison of the concentrations measured in air at delivery with the medians of concentrations measured during the European project OFFICAIR for site A and the medians of the annual average concentrations in dwellings of energy-efficient buildings of the OQAI's database for site B

VOCs	This project		European project OFFICAIR		This project			OQAI's French database
	A0	A1	Summer	Winter	B0	B1	B2	
Formaldehyde	18.3	35.9	14	7.5	<2	4.5	4.8	17.2
Acetaldehyde	5.6	8.6	6.1	4.5	4	9.4	8.8	11.5
Propionaldehyde	<1.8	<1.9	2.4	1.2	<2	2.2	—	—
Benzaldehyde	<1.8	<1.9	0.9	<LOQ	3.7	11.8	10.9	—
Hexaldehyde	7.4	8.7	10	4.4	11.9	44.3	38.2	21.3
1,4-Dichlorobenzene	<0.2	<0.21	—	—	<0.2	<0.2	<0.2	<LOD
2-Butoxyethanol	6.25	5.88	2.5	0.4	<5.6	<5.9	<5.5	1.4
n-Decane	<0.2	<0.21	—	—	0.9	0.2	8.2	2.3
Trichloroethylene	<0.2	<0.21	<LOD	<LOD	<0.2	<0.2	<0.2	<LOD
Tetrachloroethylene	<0.2	<0.21	<LOD	<LOD	<0.2	<0.2	<0.2	<LOD
Benzene	<1.26	<1.30	1	1.7	<1.39	<1.48	<1.37	1.5
Toluene	<0.2	0.98	4.7	3.1	<0.2	<0.2	1.8	4.8
Ethylbenzene	<0.2	<0.21	1.1	1	<0.2	<0.2	1.3	1.2
(m,p)-Xylenes	<0.4	<0.41	2.5	2.2	0.8	<0.5	5.9	2.7
o-Xylene	<0.2	<0.21	—	—	0.6	<0.2	3.4	1.2
Styrene	<0.2	<0.21	0.9	0.5	15.1	13.5	>22	1.1
Limonene	>20.20	3.52	—	—	0.6	<0.2	4.8	24.2

Abbreviations: “—”, data not available; LOD, limit of detection; LOQ, limit of quantification; OQAI, observatory on indoor air quality; VOCs, volatile organic compounds.

building was associated with the incidence, and exacerbations of occupational asthma for adults and visible mold and mold odor were associated with the development and exacerbations of asthma in children.²¹ Moreover, according to Caillaud et al²¹, mold exposure is associated with allergic rhinitis.

In a recent study, Reboux *et al* analyze indoor fungi by cultures of airborne samples from 1012 dwellings.⁷⁶ 908 patients suffering from rhinitis, conjunctivitis, and asthma were compared to 104 controls free of allergies. Reboux *et al* stated that a threshold of 495 UFC/m³ of *Cladosporium* is able to discriminate 90% of asthmatics. This value is almost reached at different stages suggesting the need of intervention to limit the humidity to avoid development of molds: respect of drying times, add ventilation during the construction stage, protect moisture-sensitive materials (plasterboard, insulation, wood panels, interior wood joinery).⁶⁶

In France, Ginestet *et al* showed fungal contamination in two buildings in use that can be considered low (<100 CFU/m³).⁷⁷ Nevertheless, the authors considered that the very high rate of air exchange inside the buildings may explain these low values. This is in accordance with the BASE study carried out in the United States on 100 offices buildings showing the presence of spores (on average 274 sp/m³) and fungi (on average 99 CFU/m³) in the indoor environment of office buildings where no known health problems have been demonstrated.⁷⁸

7 | CONCLUSION

At delivery, VOCs, SVOCs, and mold were detected, including one compound banned on construction products and two compounds with limited emission rates.

- Low concentrations of VOCs were measured at delivery, in relation to the use of low emissive materials (class A+): Regarding VOCs, the construction phase involves but does not lead to exceedances of the guide values established to date, but one value exceeded the future regulatory value applicable in 2023.
- 11 PAHs, 6 phthalates, permethrin, and triclosan were measured at delivery. There is no exposure limit or emission control on construction sites for SVOCs despite the endocrine disruptive effect of some of these molecules.
- Regarding mold concentrations and humidity rates, the increase in building insulation leads to an increase in humidity rates that can be related to the increased incidence of respiratory diseases, highlighting the need for moisture management measures on construction sites.

Contamination during construction can be linked to all the process implementation: choice of materials, transportation and storage (impact of weather conditions), construction processes

and assembly of the various products, sequence of tasks, and construction sequencing. These results need to be confirmed by more measurements to obtain a better estimate of the distribution of concentrations. Finally, management measures must be implemented in the same time as energy-saving measures to limit the impact of SVOC concentrations or mold growth on indoor air quality and more research must be conducted to assess cumulative exposure to these pollutants and the risk for health of the future occupants.

ACKNOWLEDGMENTS

This project was supported by the French Environment and Energy Management Agency (ADEME) (1504C0005). We thank Julien Gastine for sampling, and we thank the Environment and Health Research Laboratory (LERES) for the analysis of samples and their contributions to this work. We also thank Nicolas Naulleau and Aurélie Béroutiaux (INTECO society) for the selection and follow-up of construction sites.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

How to cite this article: Gallon V, Le Cann P, Sanchez M, Dematteo C, Le Bot B. Emissions of VOCs, SVOCs, and mold during the construction process: Contribution to indoor air quality and future occupants' exposure. *Indoor Air*. 2020;30:691–710. <https://doi.org/10.1111/ina.12647>